

Can We Predict Which Startups Win?

54,294 startups · 3 outcomes · 5 notebooks

The Journey

```
flowchart LR
    A["📊 EDA 01"] --> B["🔍 Research Questions 02"]
    B --> C["⚙️ Preprocessing 03"]
    C --> D["⚖️ Imbalance Handling 04"]
    D --> E["🛠️ Feature Selection & Modeling 05"]
```

Each notebook builds directly on the one before — decisions made in 01_EDA shaped the pipeline in 03_Preprocessing , and the resampling choices in 04 carried into every experiment in 05 .

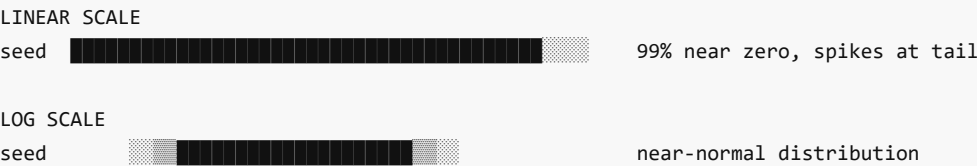
Phase 1 — EDA

01_EDA.ipynb

The raw data had widespread nulls and no ready-to-use features. Before writing a single line of preprocessing code, we mapped every missing value pattern and decided how to handle it — drop, impute, or encode as "Unknown." At the same time, we identified which raw columns could be transformed into informative features: binary round flags, log-scaled funding amounts, geographic clusters, and target-encoded categories.

```
pie title Startup Outcomes
"Operating" : 86.5
"Acquired" : 8.1
"Closed" : 5.4
```

Funding amounts — before and after log transform:



Null & Feature Engineering Decisions

#	Problem Found	Decision Made
1	15K+ rows missing founding date	Median imputation per column (year / month / quarter)
2	Geography dominated by USA	Silicon Valley, NY, Boston, Seattle, LA → 7 named clusters
3	753 markets, 115 countries	One-hot encoding impractical → target encoding (mean acquisition rate)

4	funding_total_usd VS rounds_sum mismatch	Cross-referenced to separate true zeros from missing data
5	Round amount columns mostly zeros	Converted to binary flags: has_round_A / B / C
6	Rounds D–H each below 3% participation	Too sparse individually → collapsed to round_D_plus

Phase 2 — Research Questions

02_Research_Questions.ipynb

We started with open, curious questions about what makes startups succeed — the kind you'd argue about over coffee. We then realized we could actually answer them with the data alone, no model needed.

```
flowchart TD
    RQ1["More seed\nfunding?"] --> A1["✅ 6% → 19%\nacquisition rate"]
    RQ2["Industry?"] --> A2["✅ Semiconductors\n2–3× average"]
    RQ3["Location?"] --> A3["✅ SV ~15%\nnon-USA ~3%"]
    RQ4["Angel / Grant?"] --> A4["⚠️ Angel +7pp\ngrants weak"]
    RQ5["Speed to funding?"] --> A5["✅ Under 3 months\n→ 11% acq rate"]
    RQ6["Founding era?"] --> A6["⚠️ Crisis cohort\nlower – but biased"]
```

Bias	Problem
Survivorship	Pre-2000 startups look best — failures were deleted from the dataset
Immaturity	2012+ startups not old enough to be acquired yet
Causality	Correlation only — we can't say seed funding <i>causes</i> success

Phase 3 — Preprocessing

03_Preprocessing.ipynb

Every decision from the EDA was translated into a step in a reproducible sklearn Pipeline — so the exact same transformations apply to train, validation, and test without any leakage.

```
flowchart LR
    IN["39 cols\nnulls everywhere"] --> P1["Fix dtypes"] --> P2["Parse dates"] --> P3["Fill\nunknowns"]
    P3 --> P4["Geo clusters"] --> P5["log1p funding"] --> P6["Round flags"]
    P6 --> P7["Angel flag"] --> P8["Never-funded flag"] --> P9["Days to funding"]
    P9 --> P10["Target encode"] --> P11["Median impute dates"] --> P12["Drop raw cols"]
    P12 --> OUT["46 features\n0 nulls"]
```

Split	Rows	Share
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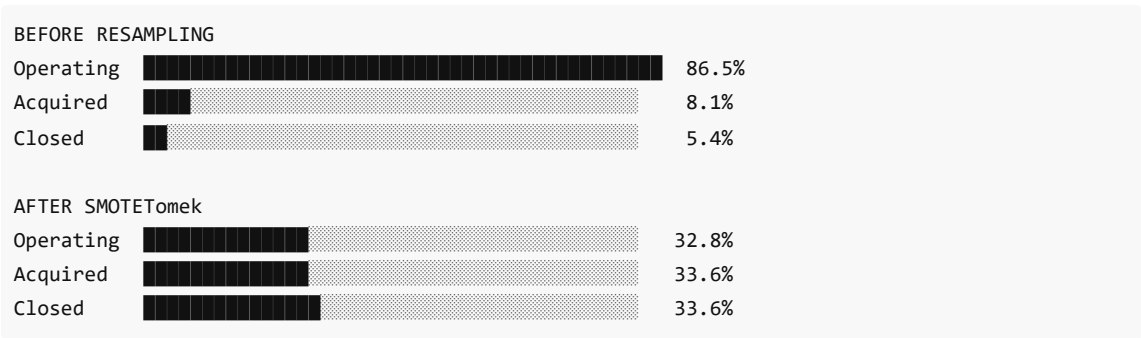
Train	27,860	70%
Validation	5,971	15%
Test	5,971	15%

Pipeline fitted on training data only — applied identically to validation and test.

Phase 4 — Imbalance Handling

04_Imbalanced_Data_Handelling.ipynb

The preprocessed training data had a 15:1 majority/minority ratio — a model with no intervention would simply learn to predict "operating" for everything. We evaluated five strategies to correct this before any model training began.



Strategy	Macro F1	Accuracy	Train Rows
Baseline (none)	0.387	86.2%	27,860
ROS	0.452	64.9%	~72,000
RUS	0.422	59.1%	~4,500
SMOTE	0.466	83.9%	62,364
SMOTETomek	0.479	83.8%	62,364
Class Weights	0.439	84.9%	27,860

All 5 variants carried forward into modeling.

Phase 5 — Feature Selection & Modeling

05_Feature_Selection_n_Modeling.ipynb

We ran a democratic feature vote across 8 methods to select the 24 most informative features, then combined them with the full experiment grid — every model against every resampling strategy — to find the best combination. The test set was used exactly once at the end.

```
flowchart LR
    F["32 features"] --> V1["ANOVA"]
    F --> V2["Chi²"]
```

```

F --> V3["Random Forest"]
F --> V4["Gradient Boosting"]
F --> V5["Lasso"]
F --> V6["Ridge"]
F --> V7["SVM"]
F --> V8["Forward Select"]
V1 & V2 & V3 & V4 & V5 & V6 & V7 & V8 --> VOTE["Vote ≥ 5/8 + corr pruning"]
VOTE --> OUT["24 features selected"]

```

The full experiment grid:

	All Features (32)	Selected Features (24)
Logistic Regression L2	× 5 resampling	× 5 resampling
Logistic Regression L1	× 5 resampling	× 5 resampling
Random Forest	× 5 resampling	× 5 resampling
Gradient Boosting	× 5 resampling	× 5 resampling
XGBoost	× 5 resampling	× 5 resampling

5 models × 2 feature sets × 5 resampling strategies = 50 experiments

Primary metric: **Macro F1** (all three classes weighted equally). Test set touched exactly once.

Key Lessons

- **Feature engineering beats model choice** — the 39 → 46 feature transformation mattered far more than which algorithm was used
- **Imbalance follows you everywhere** — resampling in 04 was only the beginning; every modeling decision downstream depended on it
- **Correlation is not causation** — we predict patterns, not mechanisms; that's an honest scope